## 7.2.5 Wrap-Up: Reflection

1. How "in the zone" do you feel right now about factoring trinomials?




# Unit 7, Lesson 3: Factoring a Trinomial With a Leading Coefficient of 1

### Benchmark

MA.912.AR.1.7 Rewrite a polynomial expression as a product of polynomials over the real number system.

### Learning Target

☐ I can factor a trinomial with a leading coefficient of $1$.

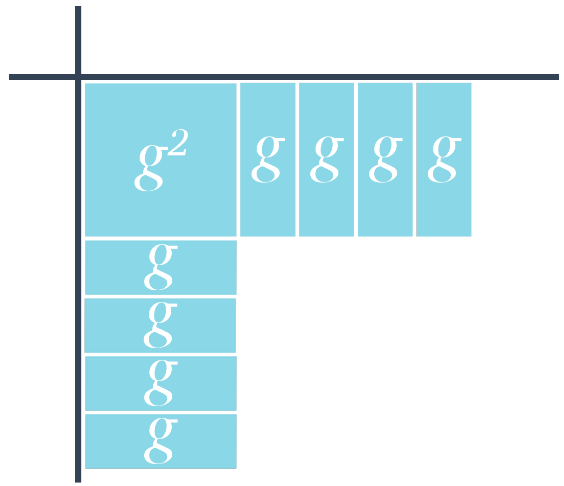
## 7.3.1 Warm-Up: Bell Work

1. Complete the table for $x^2 + 3x - 18$.

| | $x$ | ______ |
|---|---|---|
| $x$ | $x^2$ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$<br>○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$ |
| ______ | ○ $1x$ ○ $-1x$ ○ $6x$ ○ $-6x$<br>○ $2x$ ○ $-2x$ ○ $9x$ ○ $-9x$<br>○ $3x$ ○ $-3x$ ○ $18x$ ○ $-18x$ | $-18$ |

## 7.3.2 Collaboration: Different Methods of Factoring

1. Garcelle is trying to factor the trinomial $g^2 + 8g + 12$ using algebra tiles.




Discuss with your partner the mistake that Garcelle made. Write a note to Garcelle to explain her mistake and how she can fix it.

**Means**
**I**
**Start**
**To**
**Acquire**
**Knowledge**
**Experience**
**Skills**

2. Complete the table for $g^2 + 8g + 12$.

| | $g$ | ______ |
|---|---|---|
| $g$ | $g^2$ | ○ $1g$ ○ $-1g$ ○ $4g$ ○ $-4g$<br>○ $2g$ ○ $-2g$ ○ $6g$ ○ $-6g$<br>○ $3g$ ○ $-3g$ ○ $12g$ ○ $-12g$ |
| ______ | ○ $1g$ ○ $-1g$ ○ $4g$ ○ $-4g$<br>○ $2g$ ○ $-2g$ ○ $6g$ ○ $-6g$<br>○ $3g$ ○ $-3g$ ○ $12g$ ○ $-12g$ | $12$ |

3. What are the two binomial factors for $g^2 + 8g + 12$?

$$g^2 + 8g + 12 = (\qquad\qquad)(\qquad\qquad)$$

Felix showed Garcelle how to use the ***area model*** to factor $g^2 + 8g + 12$.

The ***area model*** is a rectangular diagram that utilizes the decomposition of side lengths by place value to multiply numbers using the distributive property.

4. Fill in the missing information for the ***area model*** for $g^2 + 8g + 12$.

| | $1g$ | $6$ |
|---|---|---|
| $1g$ | | $6g$ |
| $2$ | | |

5. Garcelle asked Felix how to determine the monomial that belongs in the orange boxes when creating the area model for $k^2 - 3k - 28$. Felix said that he rewrites $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$ because $-7 \cdot 4 = -28$ and $-7 + 4 = -3$. He knows that one of the orange boxes should result in the product $-7k$ and the other should result in the product $4k$.

a. Complete the area model for $k^2 - 3k - 28$.

| | ______ | ______ |
|---|---|---|
| ______ | ______ | ______ |
| ______ | ______ | ______ |

b. What are the two binomial factors for $k^2 - 3k - 28$?

$$k^2 - 3k - 28 = (\qquad)(\qquad)$$

c. Lori uses a different method of factoring. Like Felix, she rewrote $k^2 - 3k - 28$ as $k^2 - 7k + 4k - 28$, but instead paired the terms as shown.

$$(k^2 - 7k) + (4k - 28)$$

Work with your partner to finish factoring using Lori's method, known as **factoring by grouping**, using the following steps.

| $(\boldsymbol{k^2 - 7k}) + (\boldsymbol{4k - 28})$ | |
|---|---|
| Factor out the greatest common factor from each pair of terms. | $k(k-7) + \underline{\qquad}(k-7)$ |
| The binomial, $k - 7$, is common so the sum can be rewritten. | $(k-7)(k + \underline{\qquad})$ |

There are many different methods for factoring trinomials. Relating factoring to algebra tiles or the area model is useful for understanding why the factors of a trinomial of the form $ax^2 + bx + c$ are two binomials.

Using algebra tiles or the area model is also helpful in understanding the relationship the factors of the $c$-term have with the $b$-term.

6. Discuss with your partner the relationship the factors of the $c$-term have with the $b$-term based on the expressions explored in the previous problems.

## 7.3.3 Guided Instruction: Factoring Trinomials Where $a = 1$

A ***quadratic expression***, or ***quadratic equation***, is a polynomial expression or equation that contains a term of degree $2$, but no term of higher degree.

Factoring quadratic expressions in the form of trinomials such as $ax^2 + bx + c$, where $a = 1$, results in the factors $(x + g)(x + h)$ where, $g \cdot h = c$ and $g + h = b$. Later in this unit, factoring trinomials where $a$ is a value other than $1$ will be explored.

1. Consider the trinomials $y^2 + 14y + 24$, $y^2 - 14y + 24$, $y^2 + 10y - 24$, and $y^2 - 10y - 24$.

   a. Factor each trinomial.

| $y^2 + 14y + 24$ | $y^2 - 14y + 24$ | $y^2 + 10y - 24$ | $y^2 - 10y - 24$ |
|---|---|---|---|
| | | | |

   b. Describe any similarities between the four trinomials.

   c. Describe any similarities between the factored forms of the four trinomials.

2. Consider the trinomial $x^2 - 2x - 8$.

   a. Factor $x^2 - 2x - 8$ using the area model.

| | _____ | _____ |
|---|---|---|
| _____ | _____ | _____ |
| _____ | _____ | _____ |

   b. Factor $x^2 - 2x - 8$ using factoring by grouping.

   c. Discuss with your partner how to check that the two binomials of the factored form are correct.

## 7.3.4 Your Turn!

1. Factor each quadratic expression using the method of your choice.

   a. $d^2 + 11d + 24$

   b. $a^2 - 9a + 14$

   c. $x^2 + 4x - 96$

## 7.3.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson. Include something you learned and a mistake you learned from.